Linear Elasticity

- $\underline{\sigma}$ linear in $\underline{\varepsilon}$ $\rightsquigarrow$ Ψ quadratic in $\underline{\varepsilon}$
- $\dot{\underline{\sigma}} = \frac{d}{dt}\left(\frac{\partial \Psi}{\partial \underline{\varepsilon}}\right) = \underbrace{\frac{\partial^2 \Psi}{\partial \underline{\varepsilon}\, \partial \underline{\varepsilon}}}_{=: \underline{\underline{\mathbb{C}}}} : \dot{\underline{\varepsilon}}$

$\rightsquigarrow$ constant, independent from $\underline{\varepsilon}$

$\underline{\underline{\mathbb{C}}}$ Elasticity tensor

Isotropy

$$\Psi = \frac{1}{2}\lambda I_1^2 + \mu I_2 \qquad \text{(quadr. in } \underline{\varepsilon}\text{)}$$

Lamé-constants: $\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$

$$\mu = \frac{E}{2(1+\nu)}$$

$E \hat{=}$ Young's modulus, $\nu \hat{=}$ Poisson ratio

$$\underline{\sigma} = \frac{\partial \Psi}{\partial I_1}\frac{\partial I_1}{\partial \underline{\varepsilon}} + \frac{\partial \Psi}{\partial I_2}\frac{\partial I_2}{\partial \underline{\varepsilon}}$$

- $\frac{\partial I_1}{\partial \underline{\varepsilon}} = \frac{\partial \operatorname{tr} \underline{\varepsilon}}{\partial \underline{\varepsilon}} = \frac{\partial(\varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33})}{\partial \underline{\varepsilon}} = \underline{1}$
- $\frac{\partial I_2}{\partial \underline{\varepsilon}} = \frac{\partial \operatorname{tr}(\underline{\varepsilon}^2)}{\partial \underline{\varepsilon}} = ?$

(2): $\dfrac{\partial \mathrm{tr}(\underline{\varepsilon}\,\underline{\varepsilon})}{\partial \underline{\varepsilon}} = \dfrac{\partial(\underline{\varepsilon}\,\underline{\varepsilon}\cdot\underline{1})}{\partial \underline{\varepsilon}} \hat{=} \dfrac{\partial(\varepsilon_{ij}\,\varepsilon_{jk}\,\delta_{ik})}{\partial \varepsilon_{lm}}$

$= \dfrac{\partial \varepsilon_{ij}}{\partial \varepsilon_{lm}}\,\varepsilon_{jk}\,\delta_{ik} + \varepsilon_{ij}\,\dfrac{\partial \varepsilon_{jk}}{\partial \varepsilon_{lm}}\,\delta_{ik}$

$\dfrac{\partial \varepsilon_{ij}}{\partial \varepsilon_{lm}} = \begin{cases} 1 & \text{if } ij = lm \\ 0 & \text{else} \end{cases}$

$= \delta_{il}\,\delta_{jm}$

Test: $ijlm = 1212 \rightsquigarrow \delta_{11}\delta_{22} = 1$ ✓

$ijlm = 2112 \rightsquigarrow \delta_{21}\delta_{12} = 0$ ✓

$= \delta_{il}\,\delta_{jm}\,\varepsilon_{jk}\,\delta_{ik} + \varepsilon_{ij}\,\delta_{jl}\,\delta_{km}\,\delta_{ik}$

$= \varepsilon_{ml} + \varepsilon_{ml}$

$= 2\,\varepsilon_{ml}$

$\hat{=}\ 2\,\underline{\varepsilon}$

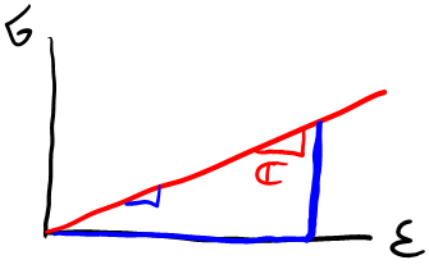
$\rightsquigarrow \underline{\sigma} = \lambda(\mathrm{tr}\,\underline{\varepsilon})\,\underline{1} + 2\mu\,\underline{\varepsilon}$ (lin. in $\underline{\varepsilon}$) ✓

$$\rightsquigarrow \underline{\underline{\mathbb{C}}} = \frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}} = \lambda \underline{1} \otimes \underline{1} + 2\mu \underline{\underline{\mathbb{I}}}$$

$$\mathbb{I}_{mnkl} = \delta_{mk}\delta_{nl}$$

with symmetry $\mathbb{C}_{ijkl} = \mathbb{C}_{klij}$

Specialty of linear elasticity:



$$\dot{\underline{\sigma}} = \underline{\underline{\mathbb{C}}} : \dot{\underline{\varepsilon}}$$

$$\rightsquigarrow \boxed{\underline{\sigma} = \underline{\underline{\mathbb{C}}} : \underline{\varepsilon}}$$

By using reduced matrix notation, this can be reformulated as

$$\underline{\underline{\sigma}} = \underline{\underline{\mathbb{C}}}\,\underline{\underline{\varepsilon}} \rightsquigarrow \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \vdots \\ \sigma_{21} \\ \vdots \end{bmatrix}_{9\times 1} = \begin{bmatrix} & & \\ & \underline{\underline{\mathbb{C}}} & \\ & & \end{bmatrix}_{9\times 9} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \vdots \end{bmatrix}_{9\times 1}$$

or by including symmetry of $\underline{\sigma}$ and $\underline{\varepsilon}$:

$$\begin{bmatrix} \sigma_{11} \\ \vdots \\ \hline \\ \end{bmatrix}_{6\times 1} = \begin{bmatrix} & \\ \underline{\underline{\mathbb{C}}} & \\ & \end{bmatrix}_{6\times 6} \begin{bmatrix} \varepsilon_{11} \\ \vdots \\ \hline 2\varepsilon_{12} \\ \vdots \end{bmatrix}_{6\times 1}$$

(Voigt notation)

$$\rightsquigarrow \underline{\underline{\mathbb{C}}} = \begin{bmatrix} \lambda+2\mu & \lambda & \lambda & & \\ \lambda & \lambda+2\mu & \lambda & \underline{\underline{0}} & \\ \lambda & \lambda & \lambda+2\mu & & \\ \hline & & & \mu & 0 & 0 \\ \underline{\underline{0}} & & & 0 & \mu & 0 \\ & & & 0 & 0 & \mu \end{bmatrix}$$

Note structure for isotropy, 3 different terms in $\underline{\underline{\mathbb{C}}}$, but still only 2 independent parameters λ, μ

with $\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$ / $\mu = \frac{E}{2(1+\nu)}$ we obtain the compliance matrix

$$\underline{\underline{\mathbb{C}}}^{-1} = \begin{bmatrix} \frac{1}{E} & -\frac{\nu}{E} & -\frac{\nu}{E} & & \\ & \frac{1}{E} & -\frac{\nu}{E} & \underline{\underline{0}} & \\ \text{sym.} & & \frac{1}{E} & & \\ \hline & & & \frac{1}{\mu} & 0 & 0 \\ & & & & \frac{1}{\mu} & 0 \\ & & & \text{sym.} & & \frac{1}{\mu} \end{bmatrix}$$

in $\underset{6\times1}{\left[\underline{\underline{\varepsilon}}\right]} = \underset{6\times6}{\left[\underline{\underline{\mathbb{C}}}^{-1}\right]} \underset{6\times1}{\left[\underline{\underline{\sigma}}\right]}$

yields e.g. $\varepsilon_{11} = \frac{1}{E}\sigma_{11} - \frac{\nu}{E}(\sigma_{22}+\sigma_{33})$

(cf. Engineering Mechanics → elastostatics)

Conditions: $\lambda > 0$, $\mu > 0$, $E > 0$, $-1 < \nu \leq 0.5$

- $\nu = 0.5 \Rightarrow \lambda \to \infty$
- $\nu > 0.5 \Rightarrow \lambda < 0$
- $\nu < 0 \rightsquigarrow$ "Metamaterial"
- $\nu < -1 \rightsquigarrow \mu < 0$

Transverse Isotropy

$$\psi = \underbrace{\frac{1}{2}\lambda I_1^2 + \mu I_2}_{\psi^{isotr}} + \alpha_1 I_1 I_4 + 2\alpha_2 I_5 + \frac{1}{2}\alpha_3 I_4^2$$

with material parameters $\lambda, \mu, \alpha_1, \alpha_2, \alpha_3$

Tensor generators

$$\frac{\partial I_1}{\partial \underline{\varepsilon}} = \underline{1} \quad , \quad \frac{\partial I_2}{\partial \underline{\varepsilon}} = 2\underline{\varepsilon} \, ,$$

$$\frac{\partial I_4}{\partial \underline{\varepsilon}} = \frac{\partial(\underline{\varepsilon}\,\underline{M}\cdot\underline{1})}{\partial \underline{\varepsilon}} \hat{=} \frac{\partial(\varepsilon_{ij} M_{jk} \delta_{ik})}{\partial \varepsilon_{lm}} = \frac{\partial \varepsilon_{ij}}{\partial \varepsilon_{lm}} M_{jk} \delta_{ik}$$

$$= \delta_{il}\delta_{jm} M_{jk} \delta_{ik}$$

$$= M_{ml}$$

$$\hat{=} \underline{M}$$

$$\frac{\partial I_5}{\partial \underline{\varepsilon}} = \frac{\partial(tr(\underline{\varepsilon}^2\underline{M}))}{\partial \underline{\varepsilon}} \hat{=} \frac{\partial(\varepsilon_{ij}\varepsilon_{jk} M_{ko}\delta_{io})}{\partial \varepsilon_{lm}}$$

$$= \frac{\partial \varepsilon_{ij}}{\partial \varepsilon_{lm}} \varepsilon_{jk} M_{ko}\delta_{io} + \varepsilon_{ij}\frac{\partial \varepsilon_{jk}}{\partial \varepsilon_{lm}} M_{ko}\delta_{io}$$

$$= \delta_{il}\delta_{jm}\varepsilon_{jk}M_{ki} + \varepsilon_{ij}\delta_{jl}\delta_{km}M_{ki}$$

$$= \varepsilon_{mk}M_{kl} + M_{mi}\varepsilon_{il}$$

$$\hat{=}\ \underline{\varepsilon}\,\underline{M} + \underline{M}\,\underline{\varepsilon}$$

Stresses:

$$\underline{\sigma} = \frac{\partial\Psi}{\partial I_1}\underline{1} + \frac{\partial\Psi}{\partial I_2}2\underline{\varepsilon} + \frac{\partial\Psi}{\partial I_4}\underline{M} + \frac{\partial\Psi}{\partial I_5}\left(\underline{\varepsilon}\,\underline{M} + \underline{M}\,\underline{\varepsilon}\right)$$

$$= \underbrace{\lambda I_1\underline{1} + 2\mu\underline{\varepsilon}}_{\sigma^{\text{isotr.}}} + \alpha_1\left(I_4\underline{1} + I_1\underline{M}\right) + 2\alpha_2\left(\underline{\varepsilon}\,\underline{M} + \underline{M}\,\underline{\varepsilon}\right) + \alpha_3\left(I_4\underline{M}\right)$$

Elasticity tensor:

$$\underline{\mathbb{C}} = \frac{\partial\underline{\sigma}}{\partial\underline{\varepsilon}} = \lambda\,\underline{1}\otimes\underline{1} + 2\mu\,\underline{\underline{\mathbb{I}}} + \alpha_1\left(\underline{1}\otimes\underline{M} + \underline{M}\otimes\underline{1}\right) + 2\alpha_2\underbrace{\left(\underline{1}\boxtimes\underline{M} + \underline{M}\boxtimes\underline{1}\right)}_{} + \alpha_3\,\underline{M}\otimes\underline{M}$$

$$\frac{\partial(\underline{\varepsilon}\,\underline{M})}{\partial\underline{\varepsilon}} \hat{=} \frac{\partial(\varepsilon_{ij}M_{jk})}{\partial\varepsilon_{lm}} = \delta_{il}\delta_{jm}M_{jk} = \delta_{il}M_{mk} = \delta_{il}M_{km} \hat{=} \left(\underline{1}\otimes\underline{M}\right)^{\overset{23}{T}}$$

Def: $\left(\underline{A}\boxtimes\underline{B}\right):\left(\underset{\sim}{a}\otimes\underset{\sim}{b}\right) = \underline{A}\,\underset{\sim}{a}\otimes\underline{B}\,\underset{\sim}{b}$

$$\rightsquigarrow A_{ij}a_jB_{kl}b_l = A_{ij}B_{kl}a_jb_l = \left(A_{ik}B_{jl}\right)^{\overset{23}{T}}a_jb_l$$

$$= \left(\underline{A}\otimes\underline{B}\right)^{\overset{23}{T}}:\underset{\sim}{a}\otimes\underset{\sim}{b}$$

Evaluation for example case $\underline{\underline{a}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$\rightsquigarrow \underline{\underline{M}} = \underline{\underline{a}} \otimes \underline{\underline{a}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\rightsquigarrow \underline{\underline{\mathbb{C}}} =$

$$\left[\begin{array}{ccc|cc} \lambda+2\mu & \lambda & \lambda+\alpha_1 & & \\ & \lambda+2\mu & \lambda+\alpha_1 & \multicolumn{2}{c}{\underline{\underline{0}}} \\ & & \lambda+2\mu+2\alpha_1+4\alpha_2+\alpha_3 & & \\ \hline & & & \mu & 0 \quad 0 \\ & \text{sym.} & & & \mu+\alpha_2 \quad 0 \\ & & & & \mu+\alpha_2 \end{array}\right]$$

Note structure ≠ isotropy,

$\rightsquigarrow$ 6 different terms, but 5 independent parameters